

LogiSim Evolution: Simulation Tool

لاڄي سيم ايڇوليشن (LogiSim Evolution) لاڄڪ سرڪٽ بنائڻ ۽ چلا ڪرڻ لاءِ ڏيکڻي ڪا سافٽ ويئر آهي.

CS XI

Chapter 1: Computer Systems

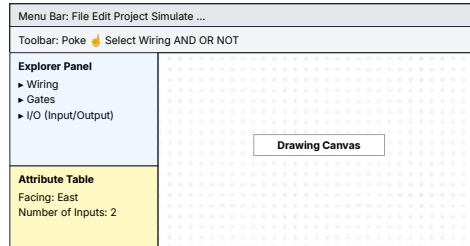
Lecture 13 of 20

LogiSim Evolution: Simulation Tool

LEARN IT

LogiSim Evolution is a free graphical tool for **designing and simulating logic circuits**. "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change. It lets you see the truth tables and Boolean expressions you learn in theory.

1. The interface areas



The main areas of the LogiSim Evolution window.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.

Wiring colours during simulation

Colour	Meaning
Bright green	Logic 1 (HIGH)
Dark green	Logic 0 (LOW)
Blue	Unknown / unconnected pin
Red	Error / short circuit

REMEMBER IT

Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is wrong".

COMMON MISTAKE

If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin. Zoom in and make sure every wire end sits exactly on a connection point.

اردو میں سمجھیں

لاجی سیم ایپولیشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔ اس میں مینو بار، ٹول بار، ایکسپلورر پینل، ایٹری بیوٹ ٹیبل اور ڈرائنگ کینوس ہوتے ہیں۔ چمکدار سبز تار 1، گہری سبز 0، نیلی تار غیر منسلک اور سرخ تار غلطی کو ظاہر کرتی ہے۔

CHECK YOURSELF

1. Where do you change the number of inputs of a gate?
2. What does a red wire mean?

Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.

Next lecture: Building a Circuit in LogiSim